26/03/2024

Phase-I
CODE-A

Aakash
Medical|IIT-JEE|Foundations

***Corporate Office*:** Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 2

Time : 3 Hrs. 20 Mins.

Answers

1. (1)	41. (4)	81. (2)	121. (3)	161. (1)
2. (2)	42. (1)	82. (2)	122. (2)	162. (3)
3. (2)	43. (2)	83. (4)	123. (1)	163. (2)
4. (3)	44. (3)	84. (4)	124. (2)	164. (1)
5. (1)	45. (3)	85. (2)	125. (2)	165. (2)
6. (1)	46. (4)	86. (4)	126. (3)	166. (4)
7. (1)	47. (3)	87. (3)	127. (3)	167. (4)
8. (3)	48. (2)	88. (4)	128. (3)	168. (4)
9. (3)	49. (2)	89. (2)	129. (2)	169. (1)
10. (4)	50. (1)	90. (3)	130. (3)	170. (4)
11. (1)	51. (2)	91. (2)	131. (2)	171. (4)
12. (1)	52. (3)	92. (2)	132. (4)	172. (1)
13. (2)	53. (3)	93. (1)	133. (1)	173. (1)
14. (3)	54. (2)	94. (3)	134. (2)	174. (4)
15. (3)	55. (1)	95. (4)	135. (4)	175. (1)
16. (3)	56. (4)	96. (3)	136. (3)	176. (1)
17. (1)	57. (3)	97. (4)	137. (2)	177. (3)
18. (2)	58. (2)	98. (4)	138. (1)	178. (3)
19. (3)	59. (4)	99. (2)	139. (2)	179. (4)
20. (4)	60. (1)	100. (1)	140. (1)	180. (2)
21. (3)	61. (2)	101. (4)	141. (2)	181. (3)
22. (3)	62. (4)	102. (3)	142. (2)	182. (2)
23. (4)	63. (3)	103. (3)	143. (4)	183. (4)
24. (1)	64. (2)	104. (1)	144. (3)	184. (3)
25. (4)	65. (3)	105. (4)	145. (1)	185. (4)
26. (2)	66. (2)	106. (1)	146. (3)	186. (1)
27. (2)	67. (2)	107. (3)	147. (2)	187. (4)
28. (1)	68. (1)	108. (2)	148. (2)	188. (4)
29. (3)	69. (3)	109. (2)	149. (2)	189. (4)
30. (2)	70. (1)	110. (3)	150. (1)	190. (4)
31. (2)	71. (2)	111. (4)	151. (1)	191. (3)
32. (1)	72. (4)	112. (1)	152. (3)	192. (4)
33. (1)	73. (2)	113. (4)	153. (3)	193. (3)
34. (4)	74. (1)	114. (3)	154. (2)	194. (2)
35. (2)	75. (3)	115. (3)	155. (2)	195. (3)
36. (3)	76. (4)	116. (2)	156. (4)	196. (1)
37. (1)	77. (4)	117. (1)	157. (2)	197. (3)
38. (1)	78. (2)	118. (4)	158. (3)	198. (4)
39. (4)	79. (2)	119. (4)	159. (4)	199. (4)
40. (1)	80. (4)	120. (4)	160. (3)	200. (2)

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FINAL TEST SERIES for NEET-2024

MM : 720 | **Test - 2** | **Time : 3 Hrs. 20 Mins.**

Answers and Solutions

PHYSICS

SECTION - A

1. Answer (1)

$$I = \Delta p = \int_0^t F\,dt \Rightarrow \int_0^t (50 - 10t)\,dt$$

$$= \int_0^t 50\,dt - \int_0^t 10t\,dt$$

$$I = \Delta p = 50t - \frac{10t^2}{2} \Rightarrow I = \Delta p = 50t - 5t^2$$

2. Answer (2)

$$a_{sys} = \frac{3g - 2g}{3 + 2} = \frac{g}{5}$$

3. Answer (2)

$$F_{net} = ma = \frac{\Delta p}{\Delta t}$$

$\Rightarrow$ Change in momentum in 1 second = 3 × 3
= 9 kg ms^{-1}

4. Answer (3)

$$Y_{com} = \frac{M_1Y_1 + M_2Y_2 + M_3Y_3 + M_4Y_4}{M_1 + M_2 + M_3 + M_4}$$

$$\Rightarrow Y_{com} = \frac{1\times 0 + 2\times \ell + 3\times \ell + 2\times 0}{8}$$

$$= \frac{2\ell + 3\ell}{8} = \frac{5\ell}{8}$$

5. Answer (1)

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} \Rightarrow \hat{i}[1-2] - \hat{j}[1-1] + \hat{k}[2-1]$$

$$\Rightarrow \vec{A}\times\vec{B} = -\hat{i} + \hat{k}$$

6. Answer (1)

Initial thrust = $m(g + a)$ = 20000(5 + 10)
= 3×10^5 N

7. Answer (1)

$$\frac{g}{5} = \frac{(m_2 - m_1)g}{m_1 + m_2} \Rightarrow m_1 + m_2 = 5m_2 - 5m_1$$

$$\Rightarrow 6m_1 = 4m_2 \Rightarrow \frac{m_1}{m_2} = \frac{4}{6} = \frac{2}{3}$$

8. Answer (3)

$T_2\cos37° = T_1\cos53°$ (from the diagram)

$$\Rightarrow T_2 \times \frac{4}{5} = T_1 \times \frac{3}{5}$$

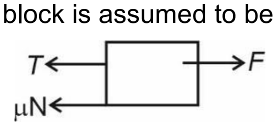
$$\Rightarrow \frac{T_1}{T_2} = \frac{4}{3}$$

9. Answer (3)

If net external force acting on the system is zero then centre of mass of system may be rest or may move with constant velocity.

10. Answer (4)

Since, the block is assumed to be at rest

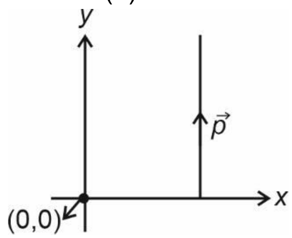


$$\Rightarrow 30 = \frac{7}{10}\times 20 + T$$

$\Rightarrow$ 30 – 14 = T

$\Rightarrow$ T = 16 N

11. Answer (1)



$|\vec{p}| = 2\times 10 = 20$ kgms^{-1} $\Rightarrow L = mvr = 20\times 2$

= 40 kgm^{2}s^{-1}

12. Answer (1)

$W = \frac{1}{2}kx^2$ and

$W_1 = \frac{1}{2}k(3x)^2$

$\Rightarrow \quad W_1 = \frac{9kx^2}{2}$

$\Rightarrow$ Additional work = $\frac{9kx^2 - kx^2}{2} = 4kx^2 = 8W$

13. Answer (2)

At equilibrium F_{cons} = 0 and $F_{cons} = -\frac{dU}{dx}$ *i.e.*

slope must be zero.

Slope is zero at $x = x_2$.

14. Answer (3)

At mean position, $Kx = mg$

$\Rightarrow x = \frac{mg}{K} = \frac{3\times 10}{1000} = 0.03$ m

$\Rightarrow$ Maximum extension = 0.06 m = 6 cm

15. Answer (3)

The correct relation are as follows:

$\vec{\tau} = \vec{r}\times\vec{F}$

$\vec{V} = \vec{\omega}\times\vec{r}$

$\vec{L} = \vec{r}\times\vec{p}$

16. Answer (3)

As body reaches maximum height, vertical displacement = $\frac{u^2\sin^2\theta}{2g}$

$\Rightarrow \quad W = mg\times\frac{u^2\sin^2\theta}{2g}\times\cos\pi = -\frac{1}{2}mu^2\sin^2\theta$

17. Answer (1)

The analogous of mass in rotational motion is moment of inertia.

18. Answer (2)

$|\vec{\tau}| = \frac{d\vec{L}}{dt} = 6t + 2$, at t = 1s

$\Rightarrow \quad |\vec{\tau}|$ = 8 N m

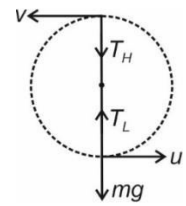
19. Answer (3)

In uniform circular motion the power delivered by the force will be zero as work done is zero.

20. Answer (4)

$T_L = mg + \frac{mu^2}{l}$

$T_H = \frac{mv^2}{l} - mg$



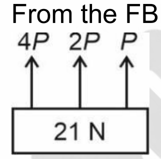
By COME, $\frac{mu^2}{2} - \frac{mv^2}{2} = mg(2l)$

$\Rightarrow \quad u^2 = v^2 + 4gl$

$\Rightarrow \quad T_L - T_H = 6mg$

21. Answer (3)

From the FBD of block,



$\Rightarrow \quad 7P$ = 21 N

$\Rightarrow \quad P$ = 3 N

22. Answer (3)

Retardation on block

$a = -\mu g$

$v = u + at$

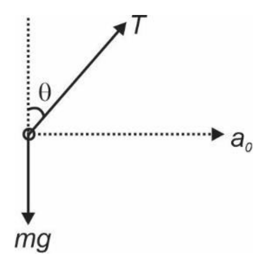
$0 = 10 + (-\mu g)t$

$\therefore \quad t = \frac{10}{\mu g} = \frac{10}{0.25\times 10} = 4$ s

23. Answer (4)

Horse moves forward due to force exerted by ground on the horse.

24. Answer (1)



$T\cos\theta = mg$

$T\sin\theta = ma_0$

$\tan\theta = \frac{a_0}{g}$

$\tan\theta = \frac{10/\sqrt{3}}{10} = \frac{1}{\sqrt{3}}$

25. Answer (4)

$$F = v_e\frac{dM}{dt} = M_0g$$

$$\therefore \quad \frac{dM}{dt} = \frac{M_0 g}{v_e}$$

$$\left(\frac{dM}{dt}\right) = \frac{6000\times 10}{1500} = 4 \times 10 = 40 \text{ kg/s}$$

26. Answer (2)

Limiting value of friction is defined for static friction.

27. Answer (2)

$$\vec{A}.\vec{B} = A.B\cos\theta$$

= 3 + 8 = 11

28. Answer (1)

Both statement are correct. Ability to do work is called energy and energy and power is scalar quantity.

29. Answer (3)

$$W = K_f - K_i$$

$$W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$= (10)^2 - (5)^2$$

= 75 J

30. Answer (2)

$$S_1 = \frac{1}{2}a(1)^2$$

$$S_2 = \frac{1}{2}a(2)^2 - \frac{1}{2}a(1)^2$$

$$= \frac{3}{2}a$$

$$S_3 = \frac{1}{2}a(3)^2 - \frac{1}{2}a(2)^2$$

$$= \frac{5}{2}a$$

$W_1 : W_2 : W_3 = S_1 : S_2 : S_3$

= 1 : 3 : 5

31. Answer (2)

Both assertion and reason are correct but reason is not correct explanation as moment of inertia depends on distribution of mass also.

32. Answer (1)

Since $\vec{F}$ is parallel to $\vec{r} \Rightarrow$ vector product is zero.

$\Rightarrow$ Ratio is zero

33. Answer (1)

In both cases acceleration should be same. In case (ii), force required is less.

34. Answer (4)

Since the hollow sphere is rotating about a fixed axis passing through it's COM hence linear velocity of its com would be zero.

35. Answer (2)

$w = Pt$

or $\frac{1}{2}mv^2 = Pt$

or $v = \sqrt{\frac{2Pt}{m}}$

$$\frac{ds}{dt} = \sqrt{\frac{2Pt}{m}}$$

$$s = \int_0^t \sqrt{\frac{2Pt}{m}}\,dt$$

or $s \propto t^{3/2}$

SECTION-B

36. Answer (3)

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

$2 \times 4 = 200 \times x^2$

$$\therefore \quad x^2 = \frac{4}{100}$$

$$\Rightarrow \quad x = \frac{2}{10}$$

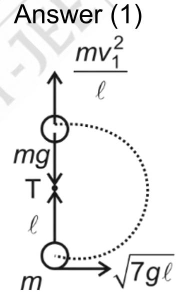
$\therefore \quad x = 20$ cm

37. Answer (1)

Using conservation of energy,

$$mg\frac{L}{2} = \frac{1}{2}\times\frac{mL^2}{3}\times\omega^2 \Rightarrow \omega = \sqrt{\frac{3g}{L}}$$

38. Answer (1)



From energy conservation

$$\frac{1}{2}mv_1^2 + mg.2\ell = \frac{1}{2}mv^2$$

$$\frac{1}{2}mv_1^2 = \frac{7}{2}mg\ell - 2mg\ell$$

$$mv_1^2 = 3mg\ell \qquad ...(1)$$

Now,

$$T = \frac{mv_1^2}{\ell} - mg = \frac{3mg\ell}{\ell} - mg$$

$$T = 2mg$$

39. Answer (4)

$\vec{A}\times\vec{B} = \vec{C} \Rightarrow$ A is perpendicular to C

40. Answer (1)

$F_{net} = nmv$

$$n = \dfrac{150}{\dfrac{50}{1000} \times 500}$$

$= 6$

41. Answer (4)

$$B\cos\theta = \frac{\vec{A}\cdot\vec{B}}{|\vec{A}|}$$

$$= \frac{6}{\sqrt{13}}$$

42. Answer (1)

$$\because\ P = \frac{dK}{dt} \Rightarrow \Delta K = \int_0^t Pdt$$

$$\Delta K = \int_0^1 (t^2 + t + 1)dt = \frac{11}{6}\ \text{J}$$

43. Answer (2)

$\omega = 6 - 3t$, at $t = 2$, $\omega = 0$

$$\Rightarrow \frac{d\theta}{dt} = 6 - 3t \Rightarrow d\theta = (6-3t)dt \Rightarrow \int d\theta$$

$$= \int_0^2 6dt - 3tdt$$

$$\theta = 6t - \frac{3t^2}{2} \Rightarrow \theta = 6\times 2 - \frac{3}{2}\times 4$$

$= 12 - 6 = 6$ rad

44. Answer (3)

$F_{net} = 20 + 0.5 \times 20 = 30$ N

Retardation $a = \dfrac{F_{net}}{m} = \dfrac{30}{2} = 15\ \text{m/s}^2$

$$t = \frac{v}{a} = \frac{10}{15} = 0.67\ \text{s}$$

45. Answer (3)

$$I_x = \frac{2}{5}MR^2 + \left(\frac{2}{5}MR^2 + M(4R^2)\right)\times 2$$

$$= \frac{2MR^2}{5} + \frac{4MR^2}{5} + 8MR^2$$

$$= \frac{6MR^2}{5} + 8MR^2 = \frac{46MR^2}{5}$$

46. Answer (4)

Inertia of body is measure of mass of body.

47. Answer (3)

$$a_c = \frac{v^2}{r} = \frac{5\times5}{10} = \frac{5}{2},\ a_t = 3\ \text{ms}^{-2}$$

$$a_{net} = \sqrt{\left(\frac{5}{2}\right)^2 + (3)^2} = 3.9\ \text{m/s}^2$$

$\Rightarrow\ F = 15.6$ N

48. Answer (2)

$$F_{cons} = \frac{-K}{r^3}$$

$$\Rightarrow \frac{K}{r^3} = \frac{mv^2}{r}$$

$$\Rightarrow mv^2 = \frac{K}{r^2}$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{K}{2r^2}$$

$$\Rightarrow \text{K.E} = \frac{K}{2a^2}$$

49. Answer (2)

If net force on rigid body is zero it may move with constant velocity.

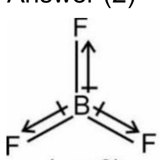
50. Answer (1)

$$P = \vec{F}.\vec{v} = Fv\cos\theta$$

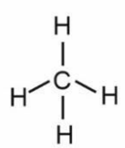
CHEMISTRY

SECTION - A

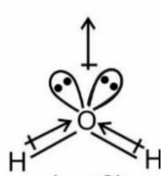
51. Answer (2)



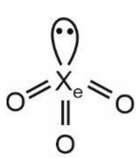
($\mu = 0$)
planar



($\mu = 0$)
non planar



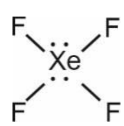
($\mu \neq 0$)
planar



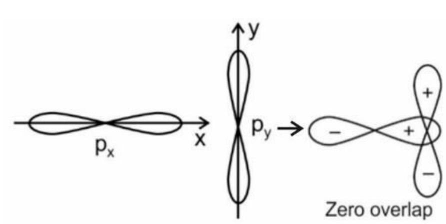
($\mu \neq 0$)
non planar

52. Answer (3)

XeF_4 has four bond pairs and two lone pair. The shape is square planar.

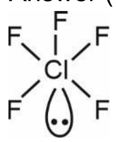


53. Answer (3)



54. Answer (2)
Carbon atom is sp^2 hybridised as it is forming 3σ and 1π bonds.

55. Answer (1)



Shape of ClF_5 is square pyramidal which contains zero 90° bonds.

56. Answer (4)

Compound	**Dipole moment (μ)**
NH_3	1.47 D
NF_3	0.23 D

57. Answer (3)
$\pi^* 2p_x$ contains two nodal planes.

58. Answer (2)
In C_2 molecule both the bonds are π bonds.

$$C_2 : (\sigma 1s)^2\ (\sigma^* 1s)^2\ (\sigma 2s)^2\ (\sigma^* 2s)^2 \begin{Bmatrix} (\pi 2p_x)^2 \\ (\pi 2p_y)^2 \end{Bmatrix}$$

59. Answer (4)
- N_2^+ and N_2^- contain odd electrons hence paramagnetic.
- O_2 contains two unpaired electrons hence paramagnetic.

60. Answer (1)

Bond order = $\dfrac{N_b - N_a}{2}$

Bond order of He_2^+ is 0.5,

$$BO(He_2) = \frac{2-1}{2} = 0.5$$

61. Answer (2)
Entropy increases with increase in disorder due to denaturation of protein on boiling of egg.

62. Answer (4)
The properties which depends upon the quantity or size of matter present in the system is called as extensive property.
Temperature, pressure and density are intensive properties.

63. Answer (3)
$\because$ For isothermal process, dt = 0

Also, total heat capacity = $\dfrac{dq}{dT} = \infty$

64. Answer (2)

$$(\text{Work done})_{Rev} = -2.303nRT \log\frac{V_f}{V_i}$$

$$= -2.303nRT \log\frac{P_i}{P_f}$$

65. Answer (3)
$q = Cm\,\Delta T$
$= 0.4 \times 21 \times (T_2 - T_1)$
$= 0.4 \times 21 \times 100$
$= 840$ J

66. Answer (2)
According to Hess's law,
a + b + c = d
$\therefore$ a + c = d – b

67. Answer (2)
Species present in their standard state has $\Delta_f H° = 0$.

68. Answer (1)
Third law of thermodynamics: The entropy of a perfectly crystalline solid at absolute zero temperature is zero.

69. Answer (3)
ΔS (total) $\neq 0$ for free expansion of an ideal gas under isothermal condition.

70. Answer (1)
$2C(s) + 2H_2(g) \rightarrow C_2H_4[\Delta H_f = 12.5 \text{ kcal}]$

$[2\Delta H_a C + 2BE_{H_2}] - [BE_{C=C} + 4BE_{C-H}] = \Delta H_f(C_2H_4)$

$2(171) + 2(104.3) - BE_{C=C} - 4(99.3) = 12.3$

$BE_{C=C} = 140.9$ kcal.

71. Answer (2)
Heat of combustion = Calorific value × Molar mass

$\Delta H_C(CH_4) \rightarrow 16x \text{ kJ mol}^{-1}$

72. Answer (4)
$\Delta G = \Delta H - T\Delta S$
At equilibrium, $\Delta G = 0$
$\Delta H - T\Delta S = 0$
$\Delta H = T\Delta S$

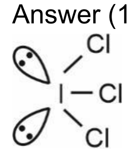
$\dfrac{\Delta H}{\Delta S} = T$

$\dfrac{170{,}000}{170} = T$

1000 K = T
i.e. T should be 1000 K.

73. Answer (2)
SF_6 : Octahedral
CCl_4 : Tetrahedral
ClF_3 : T-shape
SO_3 : Trigonal planar

74. Answer (1)

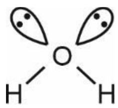


Iodine has 3 bond pairs and 2 lone pairs of electrons.
$\therefore$ Hybridisation of I = sp^3d

75. Answer (3)

Formal charge of N = 5 – 1 – 3 = +1

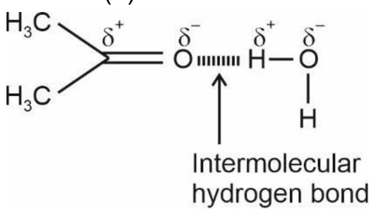
76. Answer (4)



- H_2O has exact 8 electrons in its octet.

PF_5, SF_6 and H_2SO_4 has more than 8 electrons around central atom.

77. Answer (4)



78. Answer (2)

Molecule	Bond order
F_2	1
O_2^+	2.5
N_2	3
C_2	2

79. Answer (2)

For one mole of diatomic gas γ = 1.4

80. Answer (4)

For a process to be spontaneous,

$\Delta S_{sys} + \Delta S_{surroundings} = \Delta S_{total} > 0$.

81. Answer (2)

Standard enthalpy change for the formation of 1 mole of a compound from its elements in their most stable states of aggregation is known as standard molar enthalpy of formation.

Formation of CO_2 is

$C_{(graphite)} + O_2(g) \rightarrow CO_2(g)$

82. Answer (2)

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

$$\Delta_{comb}H = \sum \Delta_f H_{(Products)} - \sum \Delta_f H_{(reactants)}$$

$$= \Delta_f H_{CO_2} + 2x\,\Delta_f H_{H_2O} - \Delta_f H_{CH_4}$$

$$= -y - 2z - (-x)$$

$$= x - y - 2z \text{ kJ/mol}$$

83. Answer (4)

$\Delta U = q + w$

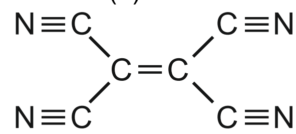
$= +100\text{ J} - 30\text{ J}$

$= 70\text{ J}$

84. Answer (4)

HF is strong acid and hydration of F^- is maximum hence it will give maximum heat of neutralisation.

85. Answer (2)



9σ bonds and 9π bonds.

SECTION-B

86. Answer (4)

Species	Bond angle
H_2O	104.5°
NH_3	107°
CH_4	109°28′
BF_3	120°

87. Answer (3)

$\Delta H = \Delta U + \Delta n_g RT$

Where $\Delta n_g \rightarrow$ difference in number of gaseous moles of products and reactants.

$\because \quad \Delta n_g = 0$

For $2HI(g) \rightleftharpoons I_2(g) + H_2(g)$

$\therefore \quad \Delta H = \Delta U$

88. Answer (4)

Applying Hess's law,

			ΔH (kJ/mol)
[A	→	B;	ΔH = – x] × 2
[2B	→	C + D;	ΔH = – y] × 3
[3C	→	E + B;	ΔH = + z] × 1
2A + 3B	→	3D + E;	ΔH = – 2x – 3y + z

89. Answer (2)

$(2p - 2p)$ axial will have maximum direction nature hence will form strongest bond.

90. Answer (3)

Higher is the enthalpy of neutralisation, stronger will be the acid.

91. Answer (2)

$$\Delta S = \frac{\Delta_{vap}H}{T} = \frac{7470}{373} = 20\text{ cal K}^{-1}\text{ mol}^{-1}$$

92. Answer (2)

Configuration of F_2 :-

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2\, \pi 2p_x^2\, \pi 2p_y^2\, \pi^* 2p_x^2\, \pi^* 2p_y^2\, \sigma^* 2p_z^0$$

Hence LUMO is σ^*2p_z

93. Answer (1)

$C(graphite) + O_2(g) \longrightarrow CO_2(g)$

When 44 g of CO_2 is formed; 400 kJ of heat is released. When 88 g of CO_2 is formed, 800 kJ of heat is released.

94. Answer (3)

$w = -P_{ex}\Delta V$

$= -5 \times (6 - 2)$

$= -5 \times 4$

$= -20$ L atm

95. Answer (4)

H-bonding takes place usually in molecules having O – H, N – H and H – F bonds.

96. Answer (3)

Polar compounds containing – OH group are easily soluble in water by H-bonding.

97. Answer (4)

$\Delta H = \Sigma(BE)_R - \Sigma(BE)_P$

$= 6(P - P) + 6(H - H) - 12(P - H)$

$= (6a + 6b - 12c)$ kJ mol^{-1}

98. Answer (4)

Bond type	**Bond length (pm)**
C – C	154
C – O	143
C – H	107
O – H	96

99. Answer (2)

$w = P\Delta V$

$= -4 \times (7.5 - 1.5)$

$= -24$ L-atm

$= -24 \times 101.3$

$= -2.4$ kJ

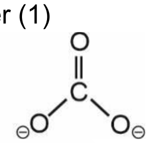
The process is adiabatic hence q = 0

$\Delta U = q + w$

$\Delta U = -2.4$ kJ

100. Answer (1)

CO_3^{2-} :



It contains $p\pi - p\pi$ bond.

BOTANY

SECTION - A

101. Answer (4)

Ground tissue in monocot stem is not differentiated into cortex and pith.

102. Answer (3)

Gymnosperms lack vessels in their xylem. They have tracheids, albuminous cells and sieve cells. They lack sieve tubes and companion cells.

103. Answer (3)

If more than two leaves arise at a node and form a whorl, it is called whorled phyllotaxy as in *Alstonia*.

104. Answer (1)

Cypsela fruit is the characteristic of Asteraceae family (Compositae).

105. Answer (4)

Cells of epiblema protrude in the form of unicellular root hairs. All tissues on the inner side of the endodermis constitute the stele.

106. Answer (1)

Sapwood is involved in water conduction. Lenticels occur in most woody trees. Phellem or cork is impervious to water due to suberin deposition.

107. Answer (3)

In cymose type of inflorescence, the main axis terminates in a flower, hence is limited in growth. The flowers are borne in a basipetal order.

108. Answer (2)

Regarding petals, imbricate aestivation is found in gulmohur, twisted aestivation in cotton, vexillary aestivation in Pea and valvate aestivation in *Calotropis*.

109. Answer (2)

In mango and coconut, fruit is known as drupe. They develop from monocarpellary superior ovaries and are one seeded. In coconut, mesocarp is fibrous.

110. Answer (3)

- Conjoint vascular bundles usually have the phloem located only on the outer side of xylem.
- All tissues except epidermis and vascular bundles constitute the ground tissue.
- In winter, cambium is less active and forms fewer xylary elements that have narrow vessels.

111. Answer (4)

Root cap covers the meristematic zone and protects it.

112. Answer (1)

Vessels are dead elements which are long, cylindrical tube-like structures made up of many cells (vessel members).

113. Answer (4)

Phylloclade has large number of spines. It is itself a modified stem.

114. Answer (3)

2-4 xylem and phloem patches are found in dicot roots whereas it is polyarch in monocot roots.

115. Answer (3)

In vexillary aestivation, the largest petal that overlaps the two smaller lateral petals is called standard.

116. Answer (2)

A mature sieve tube element possesses a peripheral cytoplasm and a large central vacuole but lacks a nucleus.

117. Answer (1)

Medicinal plants that belong to Solanaceae family are Ashwagandha and Belladonna.

118. Answer (4)

In grasses, stomata have dumb-bell shaped guard cells.

119. Answer (4)

Aleurone layer, the outermost layer of the endosperm is a triploid tissue.

120. Answer (4)

Vascular cambium of dicot roots is completely secondary in origin and wavy initially.

121. Answer (3)

In roots, protoxylem is formed towards periphery and metaxylem towards the centre, it is called exarch condition. Vascular bundles are radially arranged.

122. Answer (2)

Shoot apical meristem occupies the distant most region of the stem axis.

123. Answer (1)

Leaves are often modified to perform functions other than photosynthesis. They are converted into spines for defence as in cacti.

124. Answer (2)

Scutellum is the large shield shaped cotyledon in monocots.

125. Answer (2)

When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile as in *Citrus*.

126. Answer (3)

Mesophyll in dicot leaf is differentiated into two types of tissues palisade and spongy parenchyma. Bulliform cells are absent in dicot leaf.

127. Answer (3)

Collenchyma occurs in layers below epidermis in most of the dicotyledonous plants. Parenchyma forms major component within organs. Intercellular spaces are absent in collenchyma.

128. Answer (3)

In Australian acacia, leaves are small and short-lived.

129. Answer (2)

Gynoecium is syncarpous in lily family.

130. Answer (3)

Orchids are non-endospermic. In orchid, endosperm is not present in mature seeds and therefore called non-endospermic.

131. Answer (2)

The intercalary meristem is not a lateral meristem. These meristems are intercalated in between the mature tissues.

132. Answer (4)

Modified underground stem does not perform photosynthesis.

133. Answer (1)

Complementary cells are thin walled, redifferentiated parenchymatous cells.

134. Answer (2)

Periderm includes phellem, phellogen and phelloderm.

135. Answer (4)

Swollen placenta with many ovules are seen in plants of family Solanaceae. They have persistent calyx as in chilli, brinjal, tomato.

SECTION-B

136. Answer (3)

Adventitious roots of sweet potato get swollen and store food.

137. Answer (2)

Sclereids are commonly found in the fruit walls of nuts, seed coats of legumes and leaves of tea.

138. Answer (1)

In maize seeds, seed coat is generally fused with fruit wall. There may be variations in the length of filaments within a flower, as in *Salvia* and mustard.

139. Answer (2)

Hypodermis is collenchymatous in dicot stem. Presence of starch sheath in endodermal cells is also its characteristic feature. Water containing cavities in vascular bundles are present in monocot stem.

140. Answer (1)

Secondary growth does not occur in monocotyledons. They do not have cambium in between xylem and phloem.

141. Answer (2)

Guard cells are living and parenchymatous.

142. Answer (2)

Stilt roots – Sugarcane

Prop roots – Banyan tree

143. Answer (4)

Tyloses are balloon like outgrowth of xylem parenchyma which blocks vessels.

144. Answer (3)

The stem in plant helps in spreading out branches bearing leaves, flowers and fruits.

145. Answer (1)

Phellogen is not formed by the redifferentiation. This is a meristematic tissue that develops usually in the cortex region after dedifferentiation.

146. Answer (3)

The cells of medullary rays adjoining the intrafascicular cambium become meristematic and form the interfascicular cambium.

147. Answer (2)

Indigofera is a member of Fabaceae family having features, like flowers are zygomorphic, inflorescence is of racemose type.

148. Answer (2)

Pericycle may be thin or thick walled. In dicots it forms cambial ring for secondary growth.

149. Answer (2)

Floral formula of Solanaceae family is

$$\oplus \; ⚥ \; K_{(5)} \; \overset{\frown}{C_{(5)} A_5} \; \underline{G}_{(2)}$$

150. Answer (1)

In a dicot stem, during secondary growth primary xylem is pushed into centre.

ZOOLOGY

SECTION - A

151. Answer (1)

The sequence of amino acids *i.e.,* the positional information in a protein – which is the first amino acid, which is second, and so on – is called the primary structure of a protein.

152. Answer (3)

Paper made from plant pulp and cotton fibre is cellulosic. Cellulose is a homopolymer, which is made up of repeating glucose units.

153. Answer (3)

The substituted pyrimidine, uracil, is found only in the structure of RNA.

DNA contains adenine, thymine, guanine and cytosine.

154. Answer (2)

Dehydration involves the removal of a water molecule to link two molecules together.

Formation of guanine does not involve any dehydration reaction.

Collagen is a protein in which, the peptide bonds are formed by dehydration.

Similarly, for the formation of triglycerides, dehydration reaction is required.

155. Answer (2)

Pitch of B-DNA has 10 base pairs. So, on each step of ascent, the strand turns about 36°.

156. Answer (4)

Metal ions as co-factor form the coordination bonds with the side chains at the active site and also with the substrate molecule simultaneously.

157. Answer (2)

In cockroaches, the mouth opens into the tubular pharynx, while oesophagus opens into crop.

158. Answer (3)

A hexapeptide consists of six amino acid residues, in which the number of peptide bonds will be five only.

159. Answer (4)

Peptide bond is present between amino acids that polymerize to form insulin.

Hydrolases belong to the class III of enzymes. This enzyme catalyses hydrolysis of ester, ether, peptide, glycosidic, C – C, C-halide or P – N bonds.

160. Answer (3)

In the nucleotide sequence 5’AGACTAATG3’, there are 9 nucleotides thus, 8 phosphodiester bonds will be present.

161. Answer (1)

Oxygen makes up about 65% weight of the human body whereas carbon, hydrogen and nitrogen make up 18.5%, 0.5% and 3.3 % respectively.

162. Answer (3)

Morphine is a very effective sedative and painkiller. Morphine and codeine comes under the category of alkaloids under secondary metabolites.

163. Answer (2)

A particular property of amino acids is the ionizable nature of –COOH and $–NH_2$ groups present in it..

Hence, in solutions of different pH, the structure of amino acids changes.

164. Answer (1)

In a polysaccharide chain like glycogen, the right end is called the reducing end and the left end is called the non-reducing end.

165. Answer (2)

Starch is a homopolysaccharide that is made up of monomers of glucose.

Starch acts as the store house of energy in plant tissues.

Palmitic acid has 16 carbons including the carboxyl carbon. It is a saturated fatty acid.

166. Answer (4)

Trypsin is proteinaceous in nature whereas ribozyme is a nucleic acid that acts as enzyme.

Activity of enzymes declines both below and above the optimum temperature and pH value.

167. Answer (4)

Co-enzymes are organic compounds and their association with apoenzyme is transient, usually occurring during the course of catalysis.

168. Answer (4)

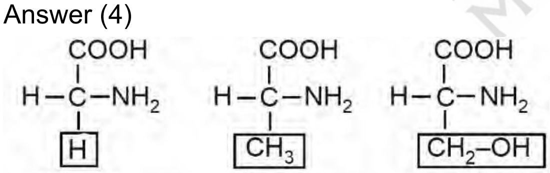
The chemical and physical properties of amino acids are essentially of the amino, carboxyl and the R functional groups. Hydroxy methyl is the functional 'R group' in serine.

169. Answer (1)

The catalytic cycle of an enzyme action can be described in the following steps:

1. First, the substrate binds to the active site of the enzyme, fitting into the active site.
2. The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate.
3. The active site of the enzyme, now in close proximity of the substrate breaks the chemical bonds of the substrate and the new enzyme-product complex is formed.
4. The enzyme releases the products of the reaction and the free enzyme is ready to bind to another molecule of the substrate and run through the catalytic cycle once again.

170. Answer (4)



Glycine Alanine Serine

A protein is imagined as a line where the first amino acid is also called the N-terminal amino acid and the last amino acid is also called the C-terminal amino acid.

So, in the given structure, the first amino acid represents glycine.

171. Answer (4)

Smooth muscles are found in the walls of internal organs such as the blood vessels, stomach and intestine.

The smooth muscle fibres taper at both ends (fusiform) and do not show striations. They are uninucleated and are unbranched.

172. Answer (1)

Skeletal muscles are striated and voluntary whereas cardiac muscles are striated and involuntary.

173. Answer (1)

The connective tissue fibres do not show conductivity. In all connective tissues, except blood and lymph, the cells secrete matrix and fibres of structural proteins such as collagen and elastin.

174. Answer (4)

Biceps is a skeletal muscle.

Skeletal muscles are voluntary in nature and possess multinucleated muscle fibres.

175. Answer (1)

Tight junctions help to stop substances from leaking across a tissue.

These junctions are found in the epithelial tissues, where they play a crucial role in creating a barrier that limits the passage of molecules and ions between adjacent cells.

Cranial nerve cells, blood and skeletal muscles (thigh muscles) do not possess tight junctions.

176. Answer (1)

Frogs have different types of sense organs, namely organs of touch (sensory papillae), taste (taste buds), smell (nasal epithelium), vision (eyes) and hearing (tympanum with internal ears). Out of these, eyes and internal ears are well-organised structures and the rest are cellular aggregations around nerve endings.

The ear is an organ of hearing as well as balancing (equilibrium).

177. Answer (3)

In frogs, a triangular structure called the sinus venosus joins the right atrium.

It receives blood through the major veins called the vena cava.

178. Answer (3)

The head of the cockroach is formed by the fusion of six segments and shows great mobility in all directions due to flexible neck.

179. Answer (4)

a.	Anal cerci in both sexes	–	Appendages arise from 10^{th} segment
b.	Testes in males	–	Present in the 4^{th}-6^{th} segments
c.	Spermatheca in females	–	Present in the 6^{th} segment
d.	Ovaries in females	–	Present in the 2^{nd} -6^{th} segments

180. Answer (2)

Body of a frog is divisible into head and trunk. A neck and tail are absent.

They do not have a constant body temperature and are called cold blooded or poikilotherms. Eyes in a frog are a pair of spherical structures situated in the orbit in the skull.

181. Answer (3)

In frogs, the final digestion of food takes place in the intestine.

182. Answer (2)

Frogs possess a 3-chambered heart and exhibit incomplete double circulation.

183. Answer (4)

Humans have closed circulatory system and they are ureotelic whereas cockroaches have open circulatory system and they are uricotelic.

184. Answer (3)

The next to last nymphal stage of cockroach has wing pads but only adult cockroaches have wings.

185. Answer (4)

The respiratory system of cockroaches consists of a network of trachea, that open through 10 pairs of spiracles present on the lateral sides of the body. Tracheal tubes (subdivided into tracheoles) carry oxygen from the air to all parts of the body. The opening of the spiracles is regulated by the sphincters. Exchange of gases takes place at the tracheoles by diffusion.

SECTION-B

186. Answer (1)

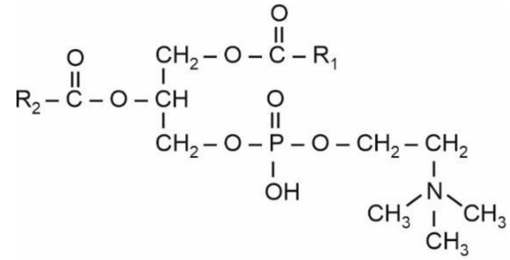
Ramachandran plot is used for studying the structure of proteins.

187. Answer (4)

Some proteins are an assembly of more than one polypeptide or subunits. The manner in which these individual folded polypeptides or subunits are arranged with respect to each other (*e.g.* linear string of spheres, spheres arranged one upon each other in the form of a cube or plate *etc.*) is the architecture of a protein otherwise called the quaternary structure of a protein. Adult human haemoglobin consists of 4 subunits. Two of these are identical to each other. Hence, two subunits of α type and two subunits of β type together constitute the human haemoglobin (Hb).

188. Answer (4)

Choline forms the polar head of lecithin and contains only one nitrogen atom.



Phospholipid (Lecithin)

189. Answer (4)

Lipids are indeed small molecular weight compounds and are present not only as such but also arranged into structures like cell membrane and other membranes. When we grind a tissue, we are disrupting the cell structure. Cell membrane and other membranes are broken into pieces, and form vesicles which are not water soluble. Therefore, these membrane fragments in the form of vesicles get separated along with the acid insoluble pool and hence in the macromolecular fraction. Lipids are not strictly macromolecules.

190. Answer (4)

1 nm = 10 Å

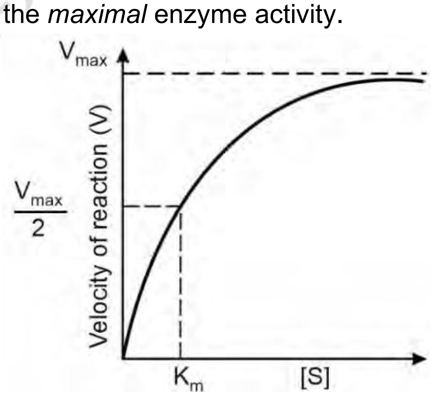
340 nm = 3400 Å

$\because$ 34 Å dsDNA has 10 base pairs

$\therefore$ 3400 Å = $\dfrac{3400\times 10}{34}$ = 1000 bp

191. Answer (3)

K_m is also known as the Michaelis-Menten constant. The value of K_m increases in the presence of a competitive inhibitor, reflecting the higher substrate concentration required for half of the *maximal* enzyme activity.



192. Answer (4)

Protein part of a holoenzyme is called apoenzyme.

193. Answer (3)

Non-essential amino acids can be made by our body, while essential amino acids are obtained from diet or food.

194. Answer (2)

Acidic amino acid – Glutamic acid

Basic amino acid – Lysine

Neutral amino acid– Valine

195. Answer (3)

The acid-insoluble fraction has only four types of organic compounds *i.e.*, proteins, nucleic acids, polysaccharides and lipids.

The molecules which have molecular weights less than one thousand dalton are usually referred to as micromolecules or simply biomolecules and they are found in the acid-soluble fraction, for example, valine and fructose.

Cellulose, DNA and insulin will be found in the acid-insoluble fraction.

196. Answer (1)

Frogs exhibit indirect development because their life cycle involves a larval stage called tadpole. Tadpole undergoes metamorphosis to form the adult.

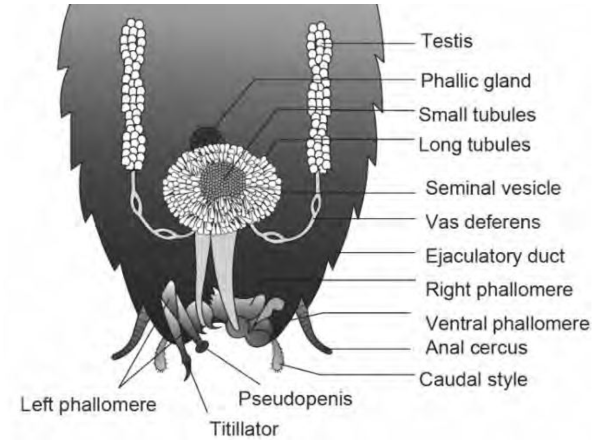
A part of frog's life cycle is spent in the water, especially during the tadpole stage. Tadpoles are aquatic and undergo significant development in water before metamorphosing into the terrestrial adult form.

197. Answer (3)

Frogs respire on land and in the water by two different methods. In water, the skin acts as the aquatic respiratory organ (cutaneous respiration).

On land, the buccal cavity, skin and lungs act as the respiratory organs.

198. Answer (4)



(The reproductive system of a male cockroach)

199. Answer (4)

Cartilage is a type of specialised connective tissue. The intercellular material of cartilage is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them. Most of the cartilages in vertebrate embryos are replaced by bones in adults. Cartilage is present in the tip of nose, outer ear joints, between adjacent bones of the vertebral column, limbs and hands in adults.

200. Answer (2)

Glycerol or trihydroxy propane is a simple lipid.

Conjugated lipids contain other chemical groups or molecules in addition to the typical fatty acid and glycerol backbone. Phospholipids (like lecithin) are conjugated lipids.

❑ ❑ ❑